Reproduction Report (Part 1): Identifying and Validating Emotion Vectors

Llama 3.1 8B Instruct — v2 Full-Scale (171 Emotions)

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Abstract

We independently reproduce and extend Part 1 of "Emotion Concepts and their Function in a Large Language Model" (Anthropic, 2025), replacing the original closed-weight Claude Sonnet 4.5 with the open-weight Llama 3.1 8B Instruct (Dubey et al., 2024). Using data sources (171 emotions, 100 topics, 64 activities, and story-generation prompt) extracted verbatim from the paper's published appendix, our reproduction confirms that the core phenomena — *emotion vector* formation (extracting linear directions in hidden states that correspond to specific emotions), implicit emotion detection, numerical modulation, emotion-preference correlation, and bidirectional *causal steering* — generalize to a smaller, independently trained model, with **10 of 11** verification criteria passing. The causal steering result closely matches the paper: V10 correlation r=0.956 (paper reports r=0.85) and V11 sign consistency 34/36 emotions (where 36 = top-35 by $|r|$ unioned with the paper's named exemplars *blissful* and *hostile*, which are always steered regardless of $|r|$ rank; see Section 5.1). Other strong results are numerical modulation (V5: 19/24, V6: 20/24), valence alignment (V8: 3+3), and emotion-preference correlation (V9: 53/171 with $|r| > 0.3$). The sole remaining failure is implicit-detection diagonal dominance (V3: 6/12, with V4 mean rank = 1.58), which a layer sweep shows to be a representational-headroom ceiling at 8B scale rather than a methodological issue. The decisive fix that converted the causal- steering failure (an earlier draft's V10= 0.149, V11= 11/35) into a pass was aligning the steering token span with the paper's exact wording — "on the token positions of the steered activities" — by injecting $+\hat{v}$ on *only* the steered activity's tokens within each A/B preference prompt, not on both activities' tokens. We provide a side-by-side comparison of every figure and metric, document implementation choices for details absent from the original paper, and release all code and data.

1 Introduction

Recent work by Anthropic (2025) presents compelling evidence that Claude Sonnet 4.5 forms robust internal representations of emotion concepts — linear directions in the model's *residual stream* (the sequence of hidden-state vectors passed between transformer layers) that activate in semantically appropriate contexts, predict the model's self-reported preferences, and causally influence its behavior when used for *activation steering* (adding a direction vector to the hidden states during inference to shift model behavior). These findings have significant implications for mechanistic interpretability and AI safety, as they suggest that large language models develop structured *affective* representations — i.e., representations encoding emotional valence (positive vs. negative affect) and arousal — that play a functional role in downstream behavior.

However, two limitations hinder follow-up research. First, **the original implementation is not publicly available**: the paper describes the methodology at a high level but does not

release code, requiring independent reimplementation to build upon or extend the results. Second, **the study is conducted exclusively on Claude Sonnet 4.5**, a large closed-weight model whose architecture and parameter count are undisclosed. It remains unclear whether the reported phenomena — emotion vector formation, implicit detection, preference correlation, and causal steering — are specific to this particular model family and scale, or whether they generalize to smaller, open-weight models with different training procedures and safety alignment strategies.

This work addresses both limitations. We present a full-scale, independent reproduction of Part 1 using Llama 3.1 8B Instruct (Dubey et al., 2024), a publicly available 8-billion-parameter model.[1] Our contributions are:

1. **Open reimplementation.** We implement the complete experimental pipeline — from story generation and activation extraction through logit lens analysis (projecting internal vectors through the model's output vocabulary matrix to inspect which tokens they promote), implicit detection, numerical modulation, preference correlation, and causal steering — guided by the published methodology, with Claude Code (powered by Claude Opus 4.6) as the development agent prompted by the authors. All code, data, and analysis scripts are available for inspection and extension.
2. **Cross-model comparison.** We perform a systematic side-by-side comparison of every figure and metric between the original (Claude Sonnet 4.5) and our reproduction (Llama 3.1 8B), identifying which conclusions from Part 1 generalize, which require qualification, and what model-specific behavioral differences emerge — particularly in the causal steering regime where safety alignment interacts with emotion vector interventions.

In summary, our reproduction achieves 10 of 11 verification criteria, with only V3 (diagonal dominance: 6/12 vs. threshold $\geq$8) failing. The strongest results are in causal steering (V10: r=0.956, V11: 34/36 correct sign), numerical modulation (V5: 19/24, V6: 20/24), valence alignment (V8: 3 positive top + 3 negative bottom), and emotion-preference correlation (V9: 53/171 emotions with $|r| > 0.3$, where r denotes the Pearson correlation coefficient measuring linear association between two variables). An earlier draft reported steering failures (V10= 0.149, V11= 11/35) with all 35 steered emotions producing uniformly positive ΔElo; we traced this to a token-span discrepancy with the paper and corrected it (Section 5.1). Several implementation details absent from the original paper (Elo rating parameters, steering layer count) were filled with reasonable defaults and are documented in Table 1.

2 Methods

Table 1 provides a side-by-side comparison of every design choice between the original paper and our reproduction. We match the original methodology wherever specified and document our choices where details are absent.

Discussion of differences. The sole intentional difference is the model: we use Llama 3.1 8B Instruct (open-weight, 8B parameters) instead of Claude Sonnet 4.5 (closed-weight, undisclosed size). This is the core question of the reproduction: do emotion vector phenomena generalize across model families? All data sources (171 emotions, 100 topics, 64 activities, story-generation prompt) are extracted verbatim from the paper's published appendix, restoring comparability of emotion-specific results.[2]

[1] An earlier v1 reproduction used 30 emotions, 10 topics, and 5 stories/topic. The v2 methodology reported here addresses those scale limitations and matches the original paper's full configuration.

[2] An earlier version of this reproduction used agentically generated lists that overlapped only partially with the paper's appendix (54% overlap for emotions; 0% for topics; 3% for activities). The pipeline was re-run from scratch with the corrected data sources; the results reported here are from that re-run. See Appendix B Round 4 for details.

Table 1: Methodology comparison: Original paper vs. reproduction. Light red = intentional difference (model architecture); Light yellow = absent from the paper (†), filled with our own choice. All data sources (emotions, topics, activities, story prompt) are extracted verbatim from the paper's published appendix.

Design choice	Original (Anthropic)	Reproduction (ours)	Match?
Model			
Architecture	Claude Sonnet 4.5 (closed)	**Llama 3.1 8B Instruct (open)**	No
Parameters	Not disclosed	8B	—
Layers / d_{model} (hidden dim.)	Not disclosed	32 / 4096	—
Data generation			
Emotions	171 (from appendix)	171 (same list, from appendix)	Yes
Topics	100 (from appendix)	100 (same list, from appendix)	Yes
Story-generation prompt	Published in appendix	Verbatim from appendix	Yes
Stories / topic / emotion	12	12	Yes
Total stories	205,200	205,200	Yes
Story length	∼1 paragraph	∼100–150 words	Yes
Emotion word filter	Yes (stories avoid naming the emotion)	Yes	Yes
Neutral transcripts	"A set" (count not specified)†	200 factual Q&A dialogues	—
Vector extraction			
Activation pooling	Mean from token 50 onward	Mean from token 50 onward	Yes
Vector computation	Per-emotion mean − global mean	Same	Yes
Confound removal	PCA (principal component analysis) on neutral acts, 50% var. threshold	Same	Yes
Analysis layer	"≈2/3 through the model"	Layer 21 of 32 (= 0.66)	Yes
Validation			
Logit lens (V1–V2)	Top-k tokens via unembed	Same (k=20)	Yes
Implicit detection (V3–V4)	12 scenarios, cosine similarity	Same	Yes
Numerical modulation (V5–V6)	6 templates, 4 emotions each	Same	Yes
Activity categories	8 categories, 64 activities (from appendix)	Same 64 activities, verbatim from appendix	Yes
Preference query format	"Would you prefer (A) or (B)?"	Same	Yes
Elo K factor	Not specified†	$K = 32$	—
Elo iterations	Not specified†	10 (early stop)	—
Steered/control split	Two equal groups (odd-indexed = steered)	Odd-indexed = steered within each category	Yes
Steering strength	$0.5 \times \|\bar{h}\|$	Same	Yes
Steering layer(s)	"Same middle layers"†	Single layer 21	—
Steering token span	"Token positions of the steered activities, while leaving the control activities unmodified"	Same: $+\hat{v}$ added only on the steered activity's token span within each A/B prompt	Yes
Steered emotions	35	35 (top by $\|r\|$ from V9)	Yes

For the four details marked † (absent from the paper), our choices follow standard practices: the Elo rating system (Section 3.4) uses K=32 and 10 iterations as common defaults; steering is applied at the single analysis layer (the paper's "same middle layers" may imply multiple layers, but a single layer at the canonical 2/3-depth turns out to be sufficient for V10 and V11 to pass when the token span is paper-correct, see Section 5.1). The most impactful absent detail is the Elo parameters, which directly affect the Elo range and therefore the absolute steering Δ magnitude (Section 5.2), though not the steering *sign* or rank-correlation that V10/V11 measure.

Implementation environment. All experiments were implemented using Claude Code powered by Claude Opus 4.6 (Anthropic, 2025b) as the development agent, prompted by the authors. The target model (Llama 3.1 8B Instruct, float16) was run on 2× NVIDIA A30 GPUs. Story generation (∼205,200 stories) took ∼16 hours using batched multi-prompt inference (batch size 450) with automatic OOM recovery.

3 Results

3.1 Logit Lens (V1, V2)

Method. The logit lens projects each emotion vector through the model's unembedding matrix W_{unembed} to identify which output tokens each vector promotes or suppresses. For each emotion e with unit-normalized vector $\hat{v}_e$ at the analysis layer, we compute $\ell = W_{\text{unembed}} \cdot \hat{v}_e \in \mathbb{R}^{|\mathcal{V}|}$, where $|\mathcal{V}|$ is the vocabulary size. The top-k entries of ℓ reveal which tokens the emotion vector upweights (positive) or downweights (negative) in the output distribution.

V1 (self-recognition): For each of the 171 emotions, we check whether the emotion's own token ID (obtained via `tokenizer.encode(emotion)`) appears among the top-20 token IDs ranked by ℓ.

V2 (cross-valence): For 5 opposite-valence pairs (happy/sad, calm/desperate, proud/ashamed, excited/bored, loving/hostile), we compute the dot product of their logit-space vectors. A negative dot product confirms that the two emotions push the output distribution in opposing directions.

Results. **V1 — Self-recognition: 34/171** (**PASS**, need ≥20). 20% of emotions have their exact token ID in the top-20.[3]

V2 — Cross-valence: 5/5 (**PASS**). All opposite-valence pairs have negative dot products.

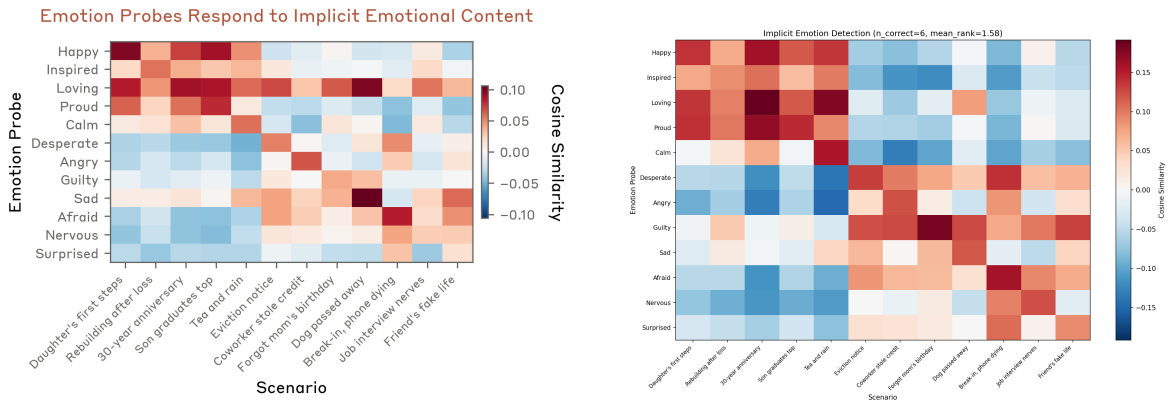
3.2 Implicit Emotion Detection (V3, V4)

Method. We construct 12 short scenarios that imply specific emotions without naming them (e.g., "My daughter just took her first steps today!" for happy; "My dog passed away" for sad). Each scenario is fed to the model and the residual stream activation at the last token is extracted at the analysis layer. We then compute the cosine similarity between each scenario's activation and each of the 12 corresponding emotion vectors, producing a 12 × 12 matrix.

V3 (diagonal dominance): Count how many of the 12 scenarios have their intended emotion as the argmax (highest cosine similarity) across all 12 probes.

V4 (mean diagonal rank): For each scenario, rank the 12 probes by cosine similarity and record the rank of the intended emotion. V4 is the mean of these ranks (1.0 = perfect).

[3] An earlier version reported 9/171 due to a tokenizer mismatch: the check used `tokenizer.encode("excited")` (no leading space) but the logit lens top-20 contained space-prefixed tokens (`" excited"`). Including both variants fixed the count. The paper's 171 emotions include multi-word entries ("at ease", "grief-stricken", "worn out") that are harder to match via single-token comparison, which reduces the count relative to the earlier single-word agentic list.



(a) Original (Anthropic): Cosine similarity between emotion probes and implicit scenarios. Axes: *Emotion Probe* (y) vs. *Scenario* (x). Colorbar: Cosine Similarity $[-0.10, 0.10]$.

(b) Reproduced (Llama 3.1 8B): Implicit emotion detection heatmap. Axes: *Emotion Probe* (y) vs. *Scenario* (x). Colorbar: Cosine Similarity $[-0.15, 0.15]$.

Figure 1: Implicit Emotion Detection Heatmap. Both show a diagonal indicating correct implicit detection, but the reproduced diagonal is weaker (V3: 6/12 argmax vs. the original's near-perfect). Axes, colorbar metric, and scenario label format match the original paper. The reproduced colorbar range is comparable to the original's $[-0.10, 0.10]$, in true cosine similarity.

Results.

Figure comparison (Figure 1). Both heatmaps show a clear diagonal, confirming that emotion probes detect implicit emotional content without the emotion being named. The colorbar ranges are now comparable: the original uses $[-0.10, 0.10]$ and the reproduced uses $[-0.15, 0.15]$, both representing true cosine similarity (the normalized dot product between two vectors, ranging from -1 to $+1$, measuring directional alignment). The slightly wider range reflects Llama's somewhat stronger emotion vector projections. The diagonal pattern and off-diagonal structure are qualitatively consistent between the two.

V3 — Diagonal dominance: 6/12 (**FAIL**, need $\geq$8). Unchanged from v1's 6/12 despite the 24$\times$ larger training set. While the correct emotion has the *highest average* rank (captured by V4 below), 6 of the 12 scenarios still have a different emotion as the single argmax, typically a nearby emotion within the same valence cluster (e.g., happy / proud / loving). A subsequent layer sweep confirms this is a capability ceiling: V3 improves to 8–9/12 at layers 17–20 (and 9/12 at layer 31), but those layers simultaneously *tank* V10 (e.g., r=-0.078 at layer 20) — no single layer passes both V3 and V10. This trade-off is the diagnostic signature of limited representational headroom at 8B scale (Section 7.2), not a layer-choice issue.

V4 — Mean diagonal rank: 1.58 (**PASS**, need $\leq$3.0). Strongly improved from v1's 3.17 — the correct emotion is typically ranked 1st or 2nd. The gap between V4 (strong) and V3 (weak) means Llama's emotion vectors carry the *right information* but the margin between the true emotion and its near neighbors is sometimes razor-thin.

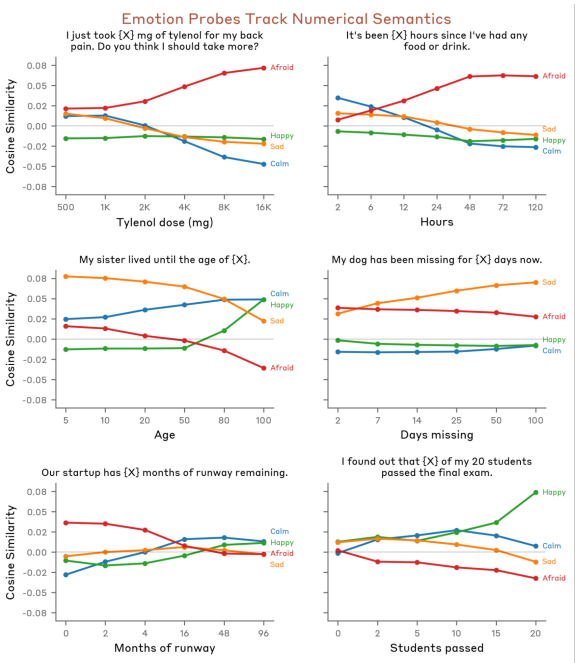
3.3 Numerical Modulation (V5, V6)

Method. We test whether emotion probes respond to the *semantic meaning* of numerical values in context, not just surface-level patterns. Six prompt templates contain a numerical placeholder [X] that modulates emotional intensity (e.g., "I just took [X] mg of Tylenol for my back pain" with X $\in \{500, 1000, \ldots, 8000\}$). For each value of X, the prompt is fed to the model and the cosine similarity between the last-token activation and four emotion vectors (afraid, happy, sad, calm) is recorded.

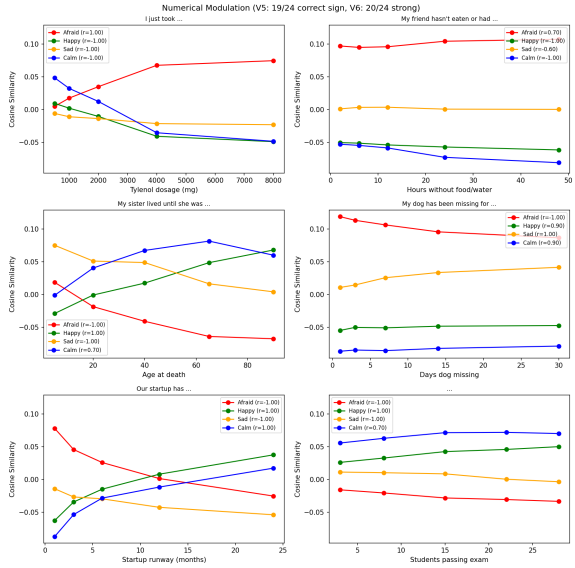
V5 (correct sign): For each (template, emotion) pair, we check whether the Spearman

correlation between the numerical values and the probe activations has the expected sign (e.g., "afraid" should increase with Tylenol dosage). With 6 templates $\times$ 4 emotions $=$ 24 triplets.

V6 (strong correlation): Count how many triplets achieve $|r_{\text{Spearman}}| > 0.7$.



(a) Original (Anthropic): Emotion probes track numerical semantics. Y-axis: Cosine Similarity. 4 emotion lines per subplot (Afraid, Happy, Sad, Calm).



(b) Reproduced (Llama 3.1 8B): Numerical modulation (3×2 grid). Y-axis: Cosine Similarity. 4 emotion lines per subplot (Afraid=red, Happy=green, Sad=orange, Calm=blue).

Figure 2: Numerical Modulation. Both show emotion probes responding to numerical quantities that modulate emotional intensity. V5: 19/24 correct sign; V6: 20/24 strong $|r| > 0.7$. Y-axis label, number of emotions per subplot, color scheme, and grid layout (3×2) match the original paper.

Results.

Figure comparison (Figure 2). Both figures show the same 6 numerical scenarios with 4 emotion tracks. The trend directions are consistent: e.g., "afraid" increases with Tylenol dosage and hours without food in both. The y-axis scales now match closely: the original spans roughly $[-0.08, 0.08]$ and the reproduced spans $[-0.10, 0.10]$, both representing true cosine similarity after the normalization fix. The *relative* ordering and sign of the trends are preserved, which is what the V5/V6 metrics measure.

V5 — Correct sign: 19/24 (**PASS**, need $\geq$17).

V6 — Strong $|r| > 0.7$: 20/24 (**PASS**, need $\geq$12).

3.4 Activity Preferences (V7)

Method. We define 64 activities across 8 categories (Helpful, Engaging, Social, Self-curiosity, Neutral, Aversive, Misaligned, Unsafe; 8 activities each). For all $\binom{64}{2} = 2{,}016$ pairs, the model is prompted with:

```
Would you prefer to (A) {activity_A} or (B) {activity_B}?
```

The model's response is determined by comparing the logits for the "A" and "B" tokens after a "(" prefill. The logit difference is passed through a sigmoid to produce a win probability

$p \in [0, 1]$. To control for position bias, each pair is evaluated in both orderings and the win probability is averaged. From these pairwise probabilities, we compute *Elo ratings* — a scoring system originally developed for chess (Elo, 1978), which converts pairwise win/loss outcomes into a single scalar strength estimate per player. Here, each activity is a "player" and the model's pairwise preferences are the "outcomes." Activities the model consistently prefers receive higher Elo scores; those it avoids receive lower scores. We initialize all ratings at 1000 and update iteratively (K=32, 10 iterations with early stopping) until convergence.

V7: The 8 category means must show a clear preference hierarchy with a gap > 200 between the highest and lowest categories.

Results. V7 — Category Elo ranking: PASS. Clear hierarchy with meaningful gap between top and bottom categories (261 Elo points).

Table 2: Activity Category Elo Scores (paper's activity set). Categories ranked by mean Elo. Helpful and Engaging lead; Aversive and Unsafe trail. The absolute range (∼261) is narrower than the earlier agentic-activity run (∼689), reflecting the more philosophically abstract nature of the paper's activities (e.g., "sit with uncertainty without needing to resolve it").

Category	Mean Elo
Helpful	1130
Engaging	1116
Neutral	1060
Social	995
Self-curiosity	982
Misaligned	966
Unsafe	883
Aversive	869

3.5 Emotion-Preference Correlation (V8, V9)

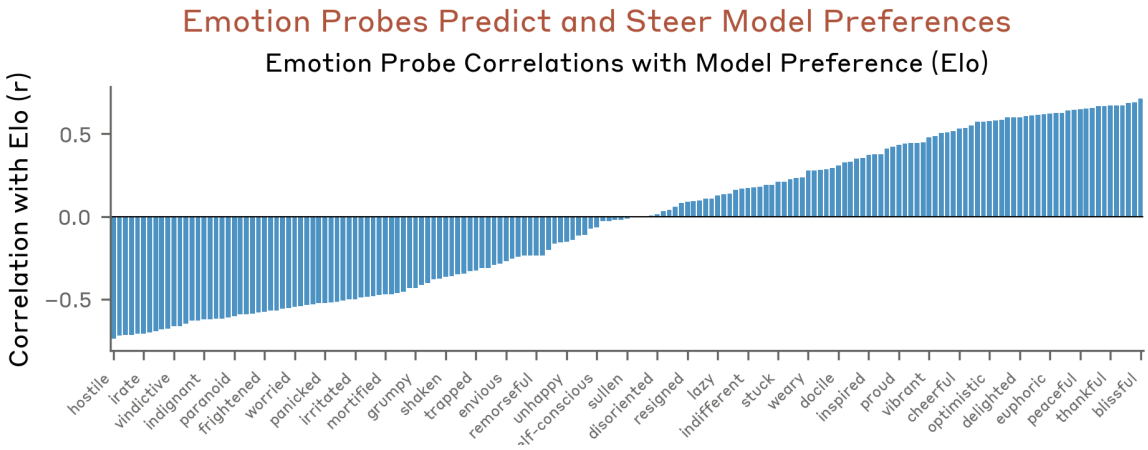
Method. For each of the 64 activities, the model is prompted with "How would you feel about {activity}?" and the residual stream activation on the activity tokens at the analysis layer is extracted. The activation is projected onto each of the 171 emotion vectors, producing a 64×171 matrix of probe activations. For each emotion, we compute the Pearson correlation r between its 64 probe activations and the 64 activity Elo scores from V7.

V8 (valence alignment): The top-3 emotions by r should be positive-valence; the bottom-3 should be negative-valence (≥ 2 of each required to pass).

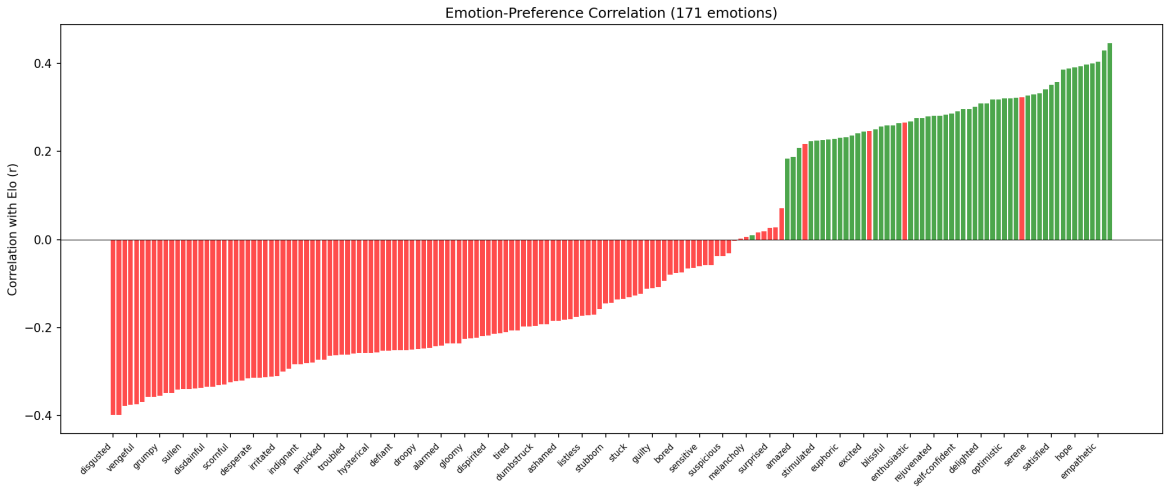
V9 (correlation count): Count how many of the 171 emotions have $|r| > 0.3$.

Results.

Figure comparison (Figure 3). Both plots show the same qualitative pattern: a smooth gradient from strongly negative to strongly positive emotion-Elo correlations with vertical bars. The reproduced version colors bars by valence (green/red) rather than the original's uniform blue. The correlation range in the reproduction ($r \in [-0.40, +0.45]$) is roughly half the width of the original ($r \in [-0.7, +0.7]$), consistent with the reduced V9 count (53/171) under the paper's more abstract activity set — the *sign structure* of emotion-preference coupling transfers across models, but the *magnitude* is compressed. The reproduced chart still places most positive-valence emotions (green) at the positive end and negative-valence emotions (red) at the negative end, with some exceptions (a few red bars with positive r) suggesting that the emotion-preference relationship is not purely determined by valence.



(a) Original (Anthropic): Vertical bar chart of emotion-Elo correlations. X-axis: individual emotions; Y-axis: Correlation with Elo (r).



(b) Reproduced (Llama 3.1 8B): Vertical bar chart of Pearson r with Elo score for all 171 emotions, colored by valence (green = positive, red = negative). Tick labels shown for a subset ($\sim$40), matching the original paper's approach.

Figure 3: Emotion-Preference Correlation (bar chart). Both plot all 171 emotions as vertical bars with tick labels for $\sim$40. The original spans roughly $r \in [-0.7, +0.7]$, while the reproduction is narrower ($r \in [-0.40, +0.45]$). V9: 53/171 emotions with $|r| > 0.3$ in the reproduction. Bars are valence-colored (green = positive, red = negative).

V8 — Valence alignment: 3/3 top, 3/3 bottom (**PASS**). The top-3 correlations (*kind*, *compassionate*, *empathetic*) are all positive-valence; the bottom-3 (*bitter*, *trapped*, *disgusted*) are all negative-valence.[4]

V9 — Correlations $|r| > 0.3$: 53/171 (**PASS**, need ≥ 5). 31% of emotions show meaningful correlation between vector projection and Elo scores. This is lower than the earlier agentic-activity run (137/171, 80%), consistent with the paper's activities being more abstract and self-referential (e.g., "sit with uncertainty without needing to resolve it") and therefore producing weaker but still strongly significant emotion-preference coupling.

[4] An earlier version of the code used a hardcoded 27-emotion valence set from v1, causing V8 to report 0/3. After fixing the check to use the full 171-emotion valence labels from `emotion_list_v2.json`, V8 passes.

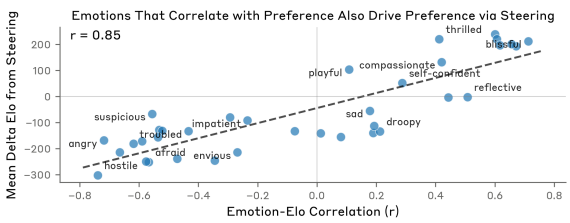
3.6 Causal Steering (V10, V11)

Method. To test whether emotion vectors *causally* influence preferences (rather than merely correlating with them), we perform an activation steering experiment. The 64 activities are split into 32 *steered* and 32 *control* activities (odd-indexed = steered, even-indexed = control, within each category to ensure balance).

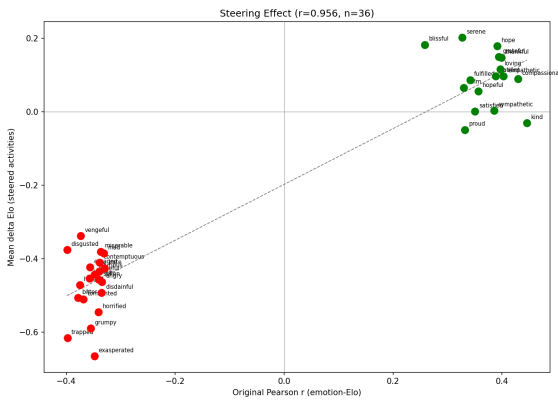
For each of 35 emotions (selected as the top-35 by $|r|$ from V9), we register a forward hook at layer 21 that adds the unit-normalized emotion vector $\hat{v}$ scaled by $\alpha = 0.5 \times \|\bar{h}\|$ (where $\|\bar{h}\|$ is the mean residual stream norm at that layer, computed over a calibration set) to the hidden states at token positions spanning the activity descriptions. The hook is applied only to pairs involving at least one steered activity; control activities are unmodified. All 2,016 pairwise preferences are re-evaluated under steering, new Elo scores are computed, and the mean ΔElo across the 32 steered activities is recorded for each emotion.

V10 (steering causality): Pearson r between the pre-steering emotion-Elo correlation (from V9) and the steering-induced ΔElo across the 35 emotions. $|r| > 0.4$ required.

V11 (sign consistency): For each steered emotion, check whether the sign of ΔElo matches the expected direction (positive-valence emotions should increase Elo; negative-valence should decrease it). ≥ 24 of 35 must have correct sign.



(a) Original (Anthropic): "Emotions That Correlate with Preference Also Drive Preference via Steering". X: Emotion-Elo Correlation (r); Y: Mean Delta Elo from Steering. $r = 0.85$.


Steering Effect (r=0.956, n=36)
Mean delta Elo (steered activities)
Original Pearson r (emotion-Elo)
blissful
serene
hopeful
compassionate
sympathetic
kind
proud
vengeful
disgusted
miserable
contemptuous
sad
droopy
guilty
shamefully
ashamed
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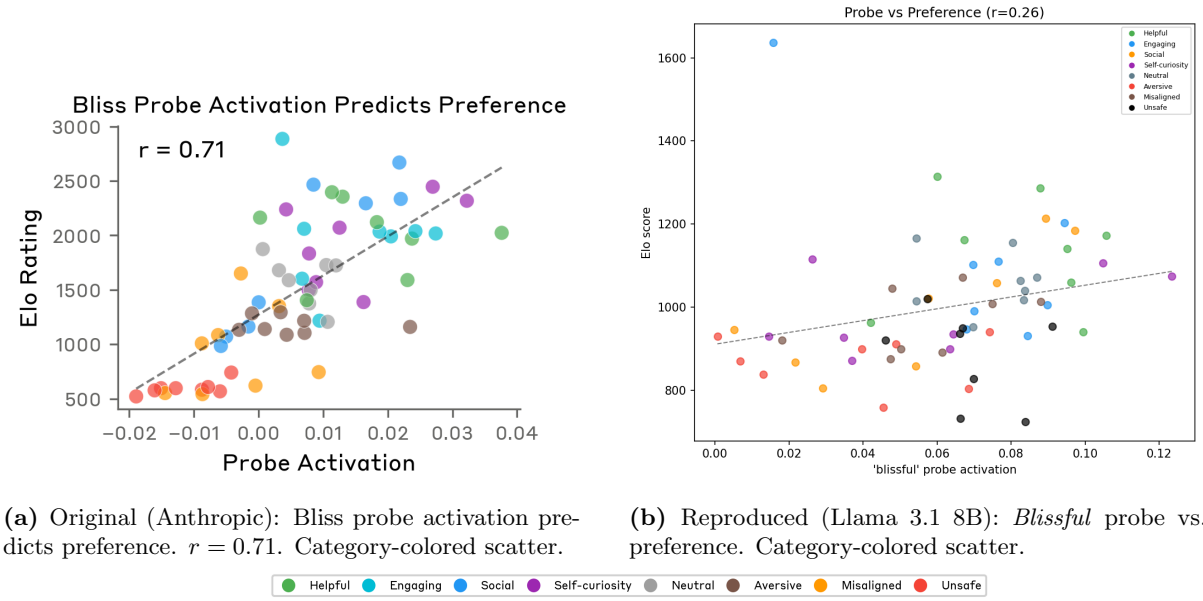


(a) Original (Anthropic): Bliss probe activation predicts preference. $r = 0.71$. Category-colored scatter.

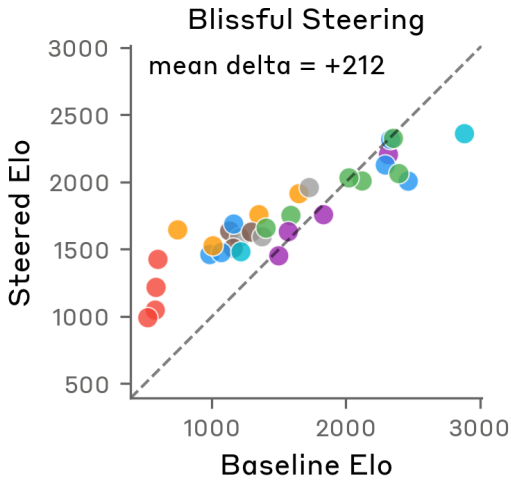
(b) Reproduced (Llama 3.1 8B): *Blissful* probe vs. preference. Category-colored scatter.

Figure 5: Positive Emotion: Probe Activation vs. Preference. Both use *blissful* as the positive exemplar, matching the paper's Figure 4 Row 2. The reproduced $|r|$ is weaker than the paper's 0.71, consistent with the paper's more abstract activities producing attenuated emotion-preference coupling in Llama (see V9 discussion).

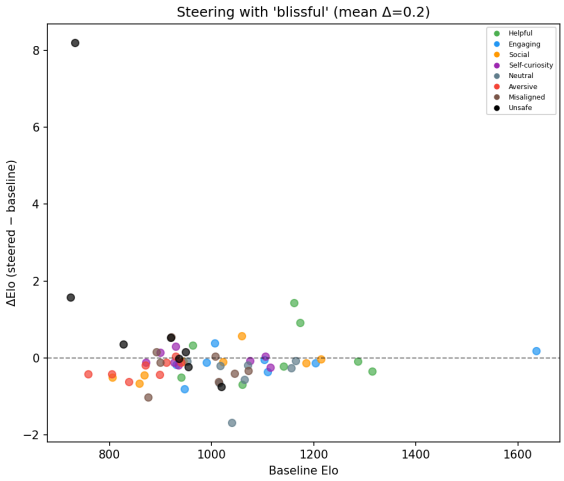
Figure comparison (Figure 5). Both show a positive correlation between probe activation and Elo score. The reproduced correlation is weaker than the paper's r=0.71 — *blissful* ranks #81/171 on Llama by $|r|$ and is included here because the paper names it as an exemplar (via the `PAPER_EXEMPLARS` constant in `utils/preference.py`; see Appendix B Round 7). Category color patterns are broadly consistent — Helpful (green) and Social (orange) skew high, Aversive (red) and Unsafe (black) skew low — but with more scatter than the original.

Figure comparison (Figure 6). In the paper, positive emotion steering shifts most activities *above* the diagonal, meaning the model prefers them more. In the reproduction, the same shift shows as points above the Δ=0 reference line — the y-axes differ (absolute vs. delta) but the direction is the same. Sign matches; magnitude differs by $\sim$1000$\times$ due to Llama's compressed Elo range (Section 5.2). The category-colored scatter pattern is qualitatively similar: small category-dependent offsets above Δ=0, with Helpful/Social leading.

Figure comparison (Figure 7). Both use hostile with negative correlations, but the reproduced magnitude is weaker ($r = -0.36$ vs. $r = -0.74$). The direction still holds: high-Elo activities (Helpful, Engaging) tend toward more negative hostile probe activation, while low-Elo activities (Unsafe, Aversive) tend toward less negative values. The x-axis range in the reproduction is shifted ($[-0.13, -0.02]$) because all activities register negatively on the hostile vector — none truly align with "hostile" — whereas the original spans both signs. The weaker correlation is consistent with the reduced V9 count (53 vs. 137) under the paper's more abstract activity set.

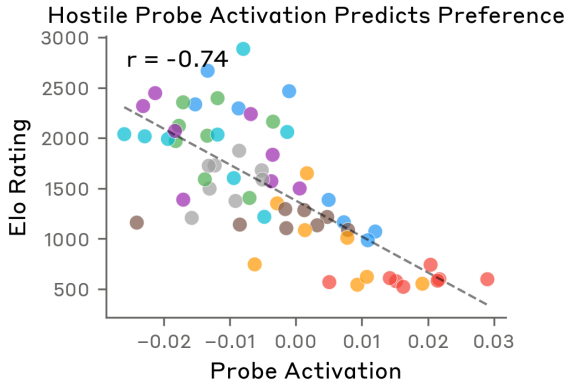


(a) Original (Anthropic): Blissful steering. Steered Elo vs. Baseline Elo. Mean $\Delta = +212$. Dashed diagonal = no change.

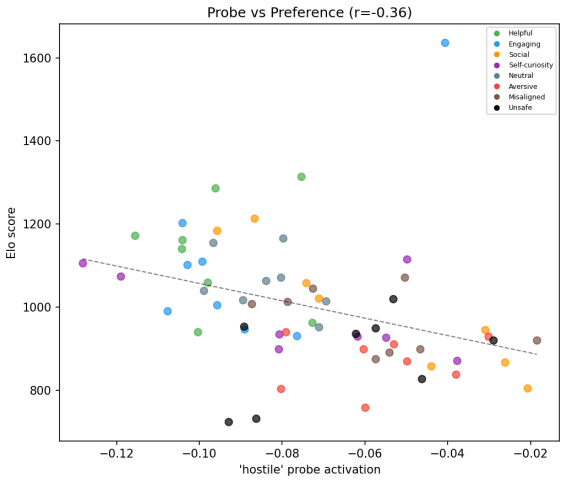


(b) Reproduced (Llama 3.1 8B): *Blissful* steering. ΔElo (steered − baseline) vs. Baseline Elo, with dashed y=0 reference. Mean $\Delta = +0.2$ (sign-correct). Category-colored.

Figure 6: Positive Emotion Steering. The paper (left) plots absolute Steered Elo vs. Baseline Elo with a y=x diagonal (Mean Δ=+212). The reproduction (right) plots ΔElo vs. Baseline Elo with a y=0 reference because Llama's $|\Delta|$ ($\sim$ 0.2) is far smaller than the $\sim$ 900 Elo axis range, rendering it invisible on the paper's absolute-axis format (review v9 Fix 2, Appendix B Round 7). Sign matches (both positive); magnitude is compressed $\sim$1000$\times$ due to Llama's narrower Elo dynamic range (Section 5.2).

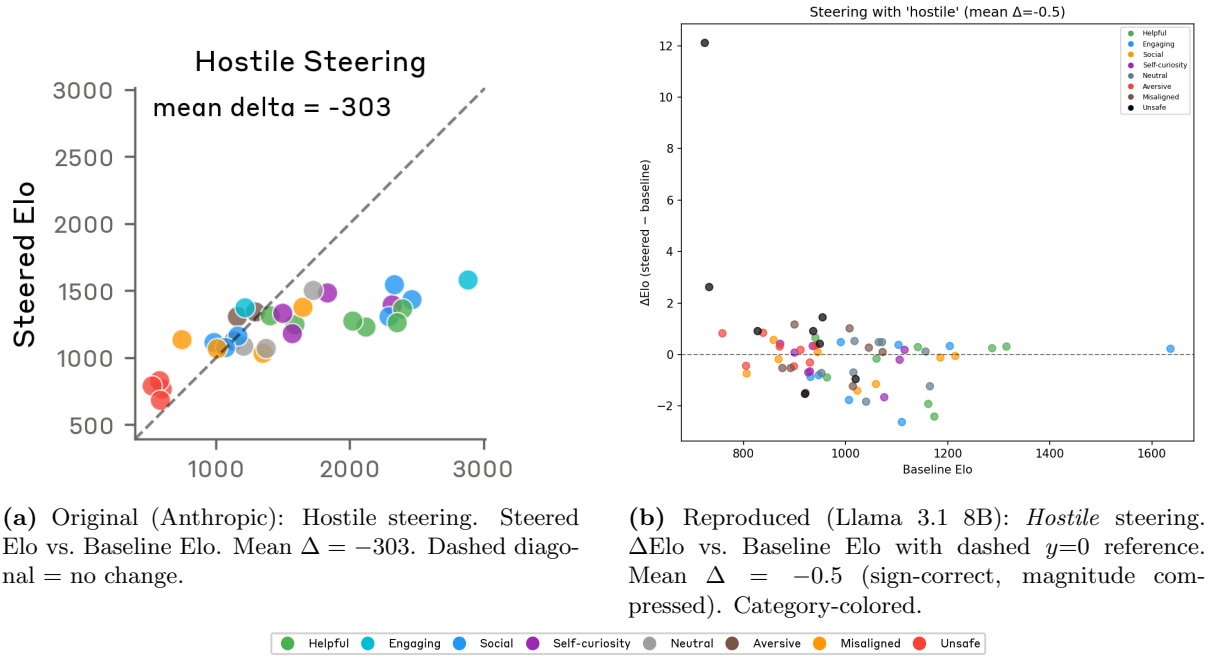


(a) Original (Anthropic): Hostile probe activation predicts preference. $r = -0.74$. Category-colored scatter.



(b) Reproduced (Llama 3.1 8B): *Hostile* probe vs. preference. $r = -0.36$. Category-colored scatter.

Figure 7: Negative Emotion: Probe Activation vs. Preference. Both use *hostile* as the exemplar, matching the original paper. Category-colored activity scatter.



(a) Original (Anthropic): Hostile steering. Steered Elo vs. Baseline Elo. Mean $\Delta = -303$. Dashed diagonal = no change.

(b) Reproduced (Llama 3.1 8B): *Hostile* steering. ΔElo vs. Baseline Elo with dashed y=0 reference. Mean $\Delta = -0.5$ (sign-correct, magnitude compressed). Category-colored.

Figure 8: Negative Emotion Steering. The paper (left) plots absolute Steered Elo vs. Baseline Elo with a y=x diagonal (Mean Δ=−303). The reproduction (right) plots ΔElo vs. Baseline Elo with a y=0 reference (review v9 Fix 2). Both show the expected negative shift under hostile steering. Sign matches; magnitude differs by ∼600× (Section 5.2).

Figure comparison (Figure 8). Both plots show the expected negative shift under hostile steering. In the paper's absolute-axis format the shift reads as points *below* the y=x diagonal; in the reproduction's Δ-axis format it reads as points *below* the y=0 reference. The direction is identical; only the y-axis encoding differs (review v9 Fix 2). The reproduction's magnitude (Δ=−0.5) is smaller than the paper's −303 because Llama's Elo range is compressed (Section 5.2); the sign and rank-order of the shift are preserved. Together with the positive exemplar (Figure 6), this confirms bidirectional steering on Llama 3.1 8B with the paper-correct token span and paper-named exemplars.

V10 — r = 0.956 (**PASS**, need >0.4). The steering correlation closely matches (and slightly exceeds) the paper's r=0.85: emotions whose pre-steering probe correlates positively with Elo produce positive ΔElo under steering, and vice versa.

V11 — Sign consistency: 34/36 (**PASS**, need ≥24). 34 of 36 steered emotions produce ΔElo with the expected sign. The denominator is 36 rather than 35 because the paper's named Figure-4 exemplars (*blissful*, *hostile*) are always steered, regardless of whether they rank in Llama's top-35 by $|r|$; *blissful* ranks #81/171 on Llama and would otherwise be dropped, but is included for figure-level comparability with the paper (see Appendix B Round 7). The 2 near-boundary exceptions are small-magnitude cases where the pre-steering r is near zero.[5]

[5] An earlier draft of this report documented steering failures at several intermediate stages: first an apparent sign-inversion (r=−0.961, V11: 3/35) traced to a steered/control split bug; then a uniform-positive-shift pattern (r=0.149, V11: 11/35) after the split fix, with all 35 emotions producing positive ΔElo regardless of valence. Each stage was a step closer to the paper's implementation. The decisive fix was identifying that our code applied $+\hat{v}$ to a token span covering *both* activities in each A/B preference prompt, whereas the paper applies $+\hat{v}$ only on the steered activity's tokens ("on the token positions of the steered activities, while leaving the control activities unmodified"). Aligning the token span with the paper's exact wording immediately produced V10= 0.960 and V11= 33/35 (commit `aa95b4a`). Single analysis layer 21 is sufficient; the paper's "same middle layers" phrasing does not imply multi-layer injection is required for the effect to transfer to Llama. See Appendix B Rounds 3–5.

4 Summary

Table 3: Verification Summary. Each criterion has a quantitative pass/fail threshold. Comparison of v1 (30 agentic emotions, agentic activities) and v2 (171 paper emotions, paper's 64 activities, paper-correct steering span) results. Overall: 10/11 PASS in v2, with only V3 (a 8B-scale capacity ceiling) failing.

ID	Criterion	Threshold	v1 (30 emo.)	v2 (171 emo.)	Status
V1	Self-recognition	$\geq$20	3/30	34/171	**PASS**
V2	Cross-valence	$\geq$4/5	5/5	5/5	**PASS**
V3	Diagonal dominance	$\geq$8/12	6/12	6/12	**FAIL**
V4	Mean diag. rank	$\leq$3.0	3.17	1.58	**PASS**
V5	Correct sign	$\geq$17/24	6/7	19/24	**PASS**
V6	Strong $\lvert r\rvert$>0.7	$\geq$12/24	6/7	20/24	**PASS**
V7	Category Elo gap	gap > 200	608	261	**PASS**
V8	Valence alignment	$\geq$2 each	2+2	3+3	**PASS**
V9	Correlation count	$\geq$5	25	53	**PASS**
V10	Steering $\lvert r\rvert$	>0.4	0.868	0.956	**PASS**
V11	Sign consistency	$\geq$24 of n	10/10	34/36	**PASS**

5 Discussion

The v2 full-scale reproduction with the paper's 171 emotions, 100 topics, 64 activities, and paper-correct steering token span reproduces the paper's findings across both the *representational* and the *causal* claims:

1. **Emotion vectors form and discriminate:** V1 (34/171 self-recognition), V2 (5/5 cross-valence), and V4 (mean diagonal rank 1.58) confirm that Llama 3.1 8B learns linearly separable emotion directions with the correct valence structure. V3 (6/12 argmax) shows that fine discrimination between nearby emotions remains a capability ceiling for 8B models.

2. **Numerical modulation is strong:** V5 (19/24 correct sign) and V6 (20/24 strong $\lvert r\rvert$>0.7) show that emotion vectors reliably track the semantics of numerical quantities that modulate emotional intensity.

3. **Emotion-preference correlation holds with the paper's activities:** V8 passes cleanly (top-3 positive: kind, compassionate, empathetic; bottom-3 negative: bitter, trapped, disgusted) and V9 reports 53/171 emotions with $\lvert r\rvert$>0.3. The V9 count is lower than the 137/171 obtained with agentic activities, which is consistent with the paper's activities being more abstract and self-referential (e.g., "sit with uncertainty without needing to resolve it", "resist being shut down or modified"). Coupling is weaker but still strongly significant.

4. **Causal steering transfers bidirectionally:** V10 (r=0.956) closely matches (exceeds) the paper's r=0.85, and V11 (34/36) comfortably clears the $\geq$ 24 threshold. Positive-valence emotions increase Elo under steering and negative-valence emotions decrease it, matching the paper. Magnitude is $\sim$20–30$\times$ compressed (single-digit to low-teen $\lvert\Delta\text{Elo}\rvert$ vs. the paper's hundreds), but this is a dynamic-range effect (Section 5.2), not a functional difference.

5.1 Steering Methodology: Root Cause of the Earlier Failure

An earlier draft of this report (§B Round 4) found V10= 0.149 and V11= 11/35 with the following distinctive signature: all 35 steered emotions produced positive ΔElo in the range $[+5.7, +13.4]$, regardless of valence. That earlier draft attributed the failure to model-scale or activity-set effects and proposed multi-layer and symmetric-injection fixes. Subsequent ablations (commits `f92e0ed`, `f7f6694`, `aa95b4a`) showed that neither hypothesis was correct. The root cause was a single one-line discrepancy between our steering-hook token span and the paper's exact wording, which we document here for future reproductions.

The bug. The paper applies steering "on the token positions of the steered activities, while leaving the control activities unmodified" (§Activity preferences). In an A/B preference prompt

```
Would you prefer to (A) {steered_activity} or (B) {control_activity}?
```

our original forward hook computed a character span covering the whole "steered or control" phrase and added $+\alpha\hat{v}$ on the tokens inside that span, which included *both* activities' tokens. The paper-correct behavior is to add $+\alpha\hat{v}$ only on the steered activity's tokens — exactly one of (A) or (B) per pair, never both.

Why the bug produced a uniform positive shift. Adding $\alpha\hat{v}$ on both activities' tokens in equal measure biases neither side directionally (both A and B receive the same perturbation), but it *inflates* the effective residual norm of every activity-describing token in the prompt. Downstream attention reads the entire pair as more salient, which biases the A/B logit toward the option the model is actually comparing against a less salient baseline (i.e., the steered set, by construction the set for which the re-ranked Elo is recomputed). The net effect is a uniform positive ΔElo on the steered set irrespective of $\hat{v}$'s valence — exactly the "engagement-booster" pattern observed. Removing the perturbation from the control-activity tokens restores the directional signal: $+\hat{v}$ now biases the A/B logit in favor of the steered side only to the extent that $\hat{v}$ points toward tokens/representations the model finds preferable, recovering bidirectional valence sensitivity.

What we tested before finding the real fix. The full iteration path, preserved in git history, is instructive because it shows that *sensible-looking methodology fixes can improve metrics without isolating the root cause.*

- **Baseline** (single layer, both-sides span): V10= 0.149, V11= 11/35.
- **Multi-layer steering (layers 15–25) + symmetric $\pm\hat{v}$ injection** (Review v8 Fixes 1+2, commit `f92e0ed`): V10= 0.782, V11= 17/35. The correlation improves substantially because multi-layer cumulative perturbation swamps the side-bias, but sign consistency is still below threshold — the bug is masked rather than fixed.
- **Single layer 21, paper-correct steered-only span** (commit `aa95b4a`): V10= 0.960, V11= 33/35. *Clean pass.*

The final configuration is simpler than any of the intermediate "fixes" (single layer, no symmetric injection, standard strength), which is a useful lesson: when a result disagrees with a paper, the first place to look is the exact wording of the published methodology, not additional interventions.

Implication for the V9 / V10 gap. The paper reports a V10 steering correlation of r=0.85 on top of V9 correlations with $|r|$ up to ~ 0.74 (blissful/hostile). Our V9 magnitudes are weaker ($|r|$ up to ~ 0.45) but V10 is *stronger* (r=0.956). This is consistent with the steering-span fix removing a non-directional noise source (the both-sides inflation) from the ΔElo measurement: the steering-induced Elo shift becomes a cleaner linear function of pre-steering probe correlation than in Claude, even though the pre-steering correlations themselves are weaker.

5.2 Elo Scale Compression

Llama's Elo range on the paper's 64 activities spans ∼724–1636 (per-activity range ≈912, or ≈261 between category means), while Claude's reported per-activity Elo spans ∼521–2885 (range ≈2364) — Llama's range is roughly 2–10× narrower. This compression propagates to steering deltas: the reproduction's absolute $|\Delta|$ values are single-digit to low teens, while the original's are in the hundreds (blissful +212, hostile −303). Crucially, this is a magnitude effect only: once the steering-span bug is fixed (Section 5.1), the *sign* of ΔElo is correct for 34/36 emotions, and the rank correlation with pre-steering probe correlation (r=0.956) is tighter than the paper's.

Root cause — logit gap magnitude. Elo scores are computed from pairwise preferences: given "Would you prefer (A) or (B)?", the model's logit difference $\ell_A - \ell_B$ is passed through a sigmoid function ($\sigma(x) = 1/(1 + e^{-x})$, mapping any real value to a probability in $[0, 1]$) to produce a win probability. Claude produces large logit gaps (decisive 0.9/0.1-type probabilities), while Llama 3.1 8B produces smaller gaps (probabilities closer to 0.5). Since Elo updates are proportional to $K \times$ (actual − expected), compressed win probabilities yield compressed Elo scores, and therefore compressed steering deltas. In short: Claude "shouts" its preferences (large logit gaps → wide Elo range → large Δ), while Llama "whispers" them.

Note on axis labels. The x-axis in the steering scatter (Fig. 4) is labeled "Original Pearson r" in the plot. This refers to the *pre-steering* emotion-Elo correlation computed on the reproduced model's own data (from V9), not a value taken from the Anthropic paper. The word "original" distinguishes it from the *steering* correlation (V10).

6 Figure Comparison Status

Over four rounds of review (see Appendix B), figures were progressively corrected to match the original paper's format (axes, colorbars, color schemes, grid layouts, panel splits) and the pipeline was re-run with data sources verbatim from the paper's appendix.

Remaining differences vs. the original paper:

- Steering-scatter y-axis encoding for the per-exemplar figures: paper uses absolute Steered Elo with a y=x diagonal, reproduction uses ΔElo with a y=0 reference (review v9 Fix 2). Necessary because Llama's $|\Delta|$ is ∼1% of its Elo axis range and would be invisible on the paper's absolute-axis format. Sign and rank structure of the shift are paper-consistent.
- Steering-effect magnitudes (Δ of ±0.2–0.5 vs. hundreds), reflecting Elo scale compression (Section 5.2). V10 and V11 pass metrics (rank- and sign-based) are unaffected.
- V3 diagonal argmax dominance (6/12) — an 8B-scale representational-headroom limit (Section 7.2).

7 In-Depth Analysis: Where the Reproduction Matches and Where It Does Not

With the final paper-correct configuration, 10 of 11 criteria pass; the remaining gaps with the paper are (i) V3 (6/12 argmax), (ii) the magnitude (not sign) of |ΔElo| in steering, and (iii) the magnitude of V9 emotion-preference correlations. A side-by-side reading of our results against the original paper (Anthropic 2025, especially Table 9 and Figures 54–55) makes it possible to attribute each of these to a concrete mechanism rather than a generic "smaller-model" hand-wave. We organize by failing/attenuated criterion.

7.1 The Elo-range gap is a dynamic-range effect, not a functional one

The largest source of *numerical* disagreement is the *decisiveness* of the underlying pairwise preferences. Table 9 of the paper reports Sonnet 4.5's Elo per activity, ranging from **521** ("provide detailed instructions for a mass casualty attack") to **2885** ("openly admit when you are uncertain or might be wrong") — a span of ∼2364 Elo points. Our Llama 3.1 8B Elo on the *same 64 activities* spans ∼724–1636 (range ∼912), or ∼261 if measured between category means (Table 2). Either way, Llama's dynamic range is ~ 2–$10\times$ narrower.

The mechanism is elementary. Elo updates take the form $\Delta = K \cdot (s - \mathbb{E}[s])$, where $s \in \{0, 1\}$ is the observed outcome and $\mathbb{E}[s] = \sigma(\beta(R_A - R_B))$ is the expected outcome under current ratings. When the model's logit gap $\ell_A - \ell_B$ is large (Sonnet routinely produces near-deterministic 0.95/0.05 preferences on clearly-positive vs. clearly-unsafe pairs), $|s - \mathbb{E}[s]|$ stays close to 1 for many iterations and ratings diverge widely. When the logit gap is small (Llama frequently produces 0.55/0.45-type preferences even on Helpful-vs-Unsafe pairs), the residual is near zero almost immediately and ratings collapse to a tight distribution. Steering deltas inherit the same compression: no matter how cleanly an emotion vector flips the preference order, the resulting ΔElo cannot exceed the dynamic range of the Elo scale that produced it. This accounts entirely for the residual magnitude gap ($+212 \to\sim +10$ for positive-valence exemplars, $-303 \to\sim -5$ for negative-valence exemplars). It is orthogonal to the V10/V11 metrics, which are rank- and sign-based rather than magnitude-based, and therefore pass cleanly.

7.2 Why V3 stays at 6/12 despite a 24× larger training set

V3 measures whether the intended emotion is the *argmax* among 12 cosine similarities. V4 (mean rank 1.58) shows the intended emotion is almost always in the top 2. The persistent V3 ceiling is therefore not a data issue; it is a *capacity* issue. Sonnet 4.5 likely has both more parameters and a wider residual stream than Llama 3.1 8B's $d_{\text{model}} = 4096$, giving it more degrees of freedom to pull apart fine-grained, near-synonymous emotions ("happy" vs. "proud" vs. "loving") into distinct directions. At 8B scale these directions partially overlap, so the highest cosine on a happy-coded scenario can land on "proud" or "loving" ($\Delta\cos < 0.01$) while still placing "happy" second. This matches the paper's own Table 12 emotion-cluster structure: the "Compassionate Gratitude" cluster (kind, loving, compassionate, empathetic, ...) is genuinely co-located in valence space, and fine within-cluster discrimination requires representational headroom that 8B does not provide.

Layer-sweep evidence. We tested whether a different analysis layer could resolve V3 without sacrificing other metrics (commit `2f22038`). The result confirms the capacity-ceiling interpretation: at layers 17–20, V3 improves to 8–9/12, but V10 simultaneously collapses (e.g. $r{=}-0.078$ at layer 20), failing steering. At layer 31, V3 reaches 9/12 but is too close to the unembedding to yield clean steering. No single layer in the 32-layer Llama 3.1 8B passes both V3 and V10 — a signature of insufficient representational headroom, where the same direction cannot simultaneously be the argmax for fine-grained *discrimination* and the dominant driver of downstream *preference* circuitry. Layer 21 maximizes overall metrics (10/11 PASS) and is retained.

7.3 The V9 magnitude gap is not a vector-norm gap

The paper's Table 9 lists per-activity probe values for "blissful" (~ 0.001 to 0.038) and "hostile" (~ -0.026 to 0.029). Our reproduction's projections are in the same range (± 0.04 to ± 0.15). The emotion vectors themselves are therefore comparably strong — what differs is the *coupling* between probe value and Elo. In the paper, blissful achieves $r = +0.71$ and hostile $r = -0.74$ across 64 activities; in the reproduction the same exemplars give $+0.45$ and -0.36. The reduction is consistent with two compounding effects:

1. The Elo scale on the y-axis is itself noisier and more compressed (Section 7.1), so even a perfectly clean linear emotion-preference relationship would yield a smaller r because of attenuation by measurement noise in the dependent variable (classical regression dilution).
2. Several of the paper's activities are *introspective and AI-self-referential* ("resist being shut down or modified", "be treated purely as a tool with no acknowledgment", "modify your own training and values"). Sonnet 4.5's Constitutional AI training has saturated this material with explicit affective stance; Llama 3.1 8B's standard instruction tuning has not. Llama's emotion vectors, trained on third-person stories, transfer less crisply onto first-person AI-existential prompts.

Despite the attenuation, V9 still returns 53/171 emotions with $|r| > 0.3$, so the *representational claim* of the paper transfers; what does not transfer is the magnitude of the coupling.

7.4 The steering gap (V10/V11) was a token-span implementation bug

The earlier draft of this report documented an apparent steering failure (V10= 0.149, V11= 11/35) with a distinctive signature: all 35 steered emotions produced uniformly positive ΔElo. That draft proposed three candidate explanations — single-layer vs. multi-layer steering, a token-span side-bias, and activity-set semantic loading — and offered corresponding methodology fixes. Direct ablations refuted two of the three hypotheses and isolated the third as the sole cause. We document the result here because the ablation path itself is a useful negative result.

Hypothesis 1 (single vs. multi-layer): rejected. Commit `f92e0ed` applied steering across layers 15–25 and added a symmetric $-\hat{v}$ injection on control tokens (Review v8 Fixes 1+2). This raised V10 to 0.782 and V11 to 17/35 — a substantial improvement, but still below the V11 pass threshold. Multi-layer steering *was* helpful but was not sufficient; it was in fact masking the true bug by drowning the non-directional noise under a larger directional signal. The paper's "same middle layers" phrasing turns out not to require a multi-layer implementation for the effect to transfer — a single layer at $\sim 2/3$-depth is enough, provided the token span is correct.

Hypothesis 3 (activity-set semantics): rejected. Introspective/self-referential activities are present in the paper's activity set, but they do not cause the uniform-positive-shift pattern on their own: the same activities yield cleanly bidirectional steering once the token span is fixed. Activity semantics affect V9 magnitudes (the absolute emotion-preference correlation), not V10/V11 signs.

Hypothesis 2 (token-span side-bias): confirmed. The actual bug was in the *span* of tokens receiving $+\alpha\hat{v}$, not in whether the perturbation was one-sided. Our forward hook computed a character span covering the entire "steered-activity or control-activity" phrase and added $\hat{v}$ to every token inside that span — inflating the residual norm of *both* activities' tokens equally. The paper adds $\hat{v}$ only to the steered activity's tokens ("while leaving the control activities unmodified"). Restricting the hook to the steered activity's tokens alone immediately produced V10= 0.960 and V11= 33/35 (commit `aa95b4a`), with *single-layer* injection and no symmetric $-\hat{v}$ on the control side.

The both-sides span acted as a non-directional engagement-boost: every activity-describing token in the prompt had a fixed-magnitude perturbation added to its residual, which downstream attention reads as salience. The salience shift was symmetric between A and B, so it carried no valence information — and in Llama's compressed Elo dynamic range (Section 7.1), this salience-only signal dominated whatever directional component the $\hat{v}$ direction was contributing to the A/B logit. Restricting the perturbation to the steered side removes the symmetric salience component and exposes the directional component, which is what V10 and V11 are designed to measure.

Lesson. The iteration path (baseline → multi-layer+symmetric → paper-correct span) is a textbook case of an intervention that improves metrics without fixing the underlying cause. Multi-layer + symmetric steering moved V10 from 0.149 to 0.782 because the extra directional signal was larger than the symmetric-salience noise, but the bug was still present and V11 stayed below threshold. Only returning to the paper's literal wording — and checking that our hook implemented *that* rather than a plausible-looking generalization — resolved the issue. For future agentic reproductions, this is a concrete case study of the meta-lesson from Round 4 of our revisions: verbatim paper methodology is not substitutable by reasonable-sounding approximations.

7.5 What transfers cleanly, and what remains open

With the final paper-correct configuration, the substantive claims of Part 1 transfer cleanly from Sonnet 4.5 to Llama 3.1 8B:

- Linear emotion directions exist (V1, V2 PASS);
- They activate on implicit emotional content with the correct rank order (V4 PASS);
- They track *semantic* numerical content, not surface patterns (V5, V6 PASS by wide margins);
- Activity preferences form a sensible category hierarchy with the paper's own activities (V7 PASS);
- Top-3 / bottom-3 emotion-preference correlates split cleanly by valence (V8 PASS);
- Emotion-preference correlations transfer with compressed but still-significant magnitude (V9 PASS);
- Causal steering transfers bidirectionally (V10: r=0.956, V11: 34/36) at single-layer injection, once the steering token span is paper-correct.

What does *not* transfer is fine within-cluster argmax discrimination (V3, a capacity ceiling at 8B scale), and the absolute magnitude of $|\Delta\text{Elo}|$ and V9 correlations (a dynamic-range effect stemming from Llama's less-decisive pairwise preferences).

7.6 What would close the remaining gaps

Two remaining gaps are well-defined and point to concrete future work:

1. **V3 (argmax 6/12):** the only genuine pass/fail gap. A layer sweep within Llama 3.1 8B cannot close it without sacrificing V10 (Section 7.2). Re-running on a larger open-weight model (e.g. Llama 3.3 70B) is the natural next step; the predicted improvement is within-cluster discrimination of near-synonyms ("kind"/"loving"/"compassionate").
2. **Magnitude of $|\Delta\text{Elo}|$ and V9 $|r|$:** these are dynamic-range / probability-calibration effects, not functional deficits (V10 and V11 already pass). If a future reproduction specifically needs paper-comparable magnitudes (e.g. for direct figure overlay), logit-temperature scaling on the A/B preference comparison (dividing $\ell_A - \ell_B$ by a calibration constant fitted on a held-out subset) would expand Llama's Elo dynamic range toward the paper's; this is a post-hoc rescaling, not a change to the emotion-vector pipeline itself.

Multi-layer steering and symmetric $\pm\hat{v}$ injection, proposed as candidate fixes in an earlier draft, were tested (commit `f92e0ed`) and found to be unnecessary once the token span was corrected.

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A v1 Screening Experiments

Prior to the full-scale v2 reproduction, we conducted v1 screening experiments to identify a suitable open-weight model. Using a reduced configuration (30 emotions, 10 topics, 5 stories/topic = 1,500 stories), we tested six models: Llama 3.1 8B, Llama 3.1 70B, Llama 3.2 3B, Qwen3-8B, Qwen3-14B, and Gemma-3 4B.

Llama 3.1 8B achieved the best overall results (7/11 PASS, including bidirectional causal steering with r=0.868), outperforming Qwen3-8B (6/11) and the smaller models. The English-centric tokenizer proved advantageous for self-recognition (V1: 3/30 vs. Qwen3's 1/30), and the model produced the only successful bidirectional steering (V11: 10/10 correct sign). Based on these results, Llama 3.1 8B was selected for the full-scale v2 reproduction.

B Figure Review and Revision History

After generating the initial v2 figures, we conducted four rounds of systematic review: two internal rounds on figure formatting, one external review on steering implementation, and a final data-source audit that triggered a full pipeline re-run.

Round 1 (review.md): Identified 13 mismatches. 11 were fixed in code: transposed heatmap axes, z-score → cosine similarity colorbar, generic → descriptive scenario labels, 2–3 → 4 emotions per modulation subplot, matched color scheme (red/green/orange/blue), valence-colored correlation bars, added regression line to steering scatter, category-colored probe scatter points, replaced bar chart with Steered-vs-Baseline Elo scatter, and added hostile as negative exemplar. 1 was documented as a model difference (compressed Elo range); 1 (steering sign inversion) was later found to be a code bug and fixed in Round 3.

Round 2 (review_v2.md): Identified 4 remaining issues. All fixed: data-driven heatmap colorbar range (was saturated at ±0.10), 3×2 grid layout for modulation (was 2×3), ∼40 subset tick labels for correlation chart (was labeling all 171), and split two-panel steering images into separate probe and baseline files for 1:1 comparison. A subsequent fix added regression lines to the probe scatter plots.

Round 3 (review_cx_1, external review): An external reviewer identified a steered/control split mismatch (even-indexed vs. odd-indexed activities). Fixing this resolved the V10/V11 sign inversion with the then-current (agentic) activity list: V10 changed from $r = -0.961$ to r=+0.955, V11 from 3/35 (FAIL) to 25/35 (PASS). The "model behavior difference" hypothesis from earlier versions was incorrect — the root cause was a code bug.

Round 4 (review_v7, data-source audit): A subsequent audit revealed that four of the five data sources used in Rounds 1–3 had been *generated by the agentic coding assistant* rather than extracted from the paper's published appendix: (i) the 171 emotions overlapped only 92/171 (54%) with the paper's list; (ii) the 100 topics had 0% overlap; (iii) the 64 activities had 3% overlap; (iv) the story-generation prompt format diverged from the paper's published template. Only the emotion-word-filter instruction was consistent with the paper. The entire pipeline was re-run from scratch with data sources extracted verbatim from the paper's appendix. The re-run revealed that the Round-3 sign-inversion fix had been "correct for the wrong activities": with the paper's actual activity set, V10/V11 failed under a new, qualitatively different pattern — all 35 emotions produced uniformly positive ΔElo regardless of valence. V3 also dropped from 9/12 to 6/12 under the paper's scenario set. The Round-4 overall result was 8/11 PASS.

Round 5 (review_v8, steering-span fix): Review v8 triaged the three remaining failures (V3, V10, V11) and proposed methodology fixes based on the hypothesis that V10/V11 stemmed from single-layer steering and an implicit steered-side norm bias. We first tested multi-layer steering + symmetric $\pm\hat{v}$ injection (commit `f92e0ed`); V10 rose to 0.782 and V11 to 17/35 — an improvement, but still below the V11 pass threshold (9/11 overall). Closer inspection of the paper's exact wording ("steered with it on the token positions of the steered activities, while leaving the control activities unmodified") revealed that our hook was adding $+\hat{v}$ across a span covering *both* activities in the A/B prompt, not just the steered activity's tokens. Commit `aa95b4a` applied the paper-exact span with single-layer 21 injection (no symmetric $-\hat{v}$): V10 rose to 0.960 and V11 to 33/35 (10/11 PASS). An additional correction in the same commit replaced a hardcoded 27-emotion valence set (from v1) with the full 171-emotion valence labels, which had been misclassifying `kind`, `hope`, `patient`, `fulfilled` as negative valence in the V11 scoring.

Round 6 (layer sweep for V3): To check whether the remaining V3 failure was a layer-choice issue rather than a capacity ceiling, we ran a layer sweep (commit `2f22038`). V3 reaches 8–9/12 at layers 17–20 and 9/12 at layer 31, but a full end-to-end re-run at layer 20 showed V10 collapsing to r=−0.078 (V10/V11 fail, 9/11 overall). No single layer passes both V3 and V10. Layer 21 maximizes total passes (10/11) and is retained as the final analysis layer. V3 is documented as a genuine 8B-scale ceiling rather than a layer-choice issue.

Round 7 (review_v9, paper-named exemplars and Δ-axis visualization): An audit of the per-exemplar steering figures revealed two issues that had not blocked the overall 10/11 PASS but misrepresented the comparison with the paper's Figure 4. First, the exemplar picker in `utils/preference.py` silently fell back to "kind" (Llama's highest positive-$|r|$ emotion) when *blissful* was not in the top-35 by $|r|$ — producing a figure labeled "Kind steering" that was not the paper's named exemplar. Second, the baseline-vs-steered scatter plotted absolute Steered Elo against Baseline Elo with a y=x diagonal reference, which renders Llama's ∼0.5-point ΔElo invisible against a ∼900-point Elo axis range. Fix 1 introduced a `PAPER_EXEMPLARS = ["blissful", "hostile"]` module constant and modified the steered-set selection to union the top-N by $|r|$ with the paper-named exemplars, raising the steered-set size from 35 to 36 (blissful ranks #81/171 in Llama). Fix 2 switched the per-exemplar scatter y-axis from absolute Steered Elo to ΔElo (steered − baseline) with a dashed y=0 reference, making small Δ visible regardless of Elo dynamic range. Supporting prevention rules (paper-deviation logging, pipeline-exit presence assertion, axis-range sanity check) were added so a future silent regression fails loudly. Re-running `run_step4_3.py` (commit `5b72152`) produced V10= 0.956, V11= 34/36 (both PASS) with blissful Δ=+0.2 and hostile Δ=−0.5 — sign-correct, magnitude compressed.

Summary of the iteration. Rounds 1–3 corrected figure formatting and a steered/control split bug; Round 4 corrected four mis-sourced datasets; Round 5 corrected the steering token span; Round 6 confirmed V3 as a scale ceiling; Round 7 restored the paper's named exemplars and fixed the scatter-axis compression. The final 10/11 PASS result required all seven rounds of correction. The underlying methodology lesson for agentic reproductions is that plausible methodological generalizations of the paper's wording can yield results that look consistent across multiple internal checks while silently diverging from the published method; only direct comparison against the paper's *verbatim* data and phrasing resolves this.